

Demonstrating User Authentication via Electrical Muscle Stimulation

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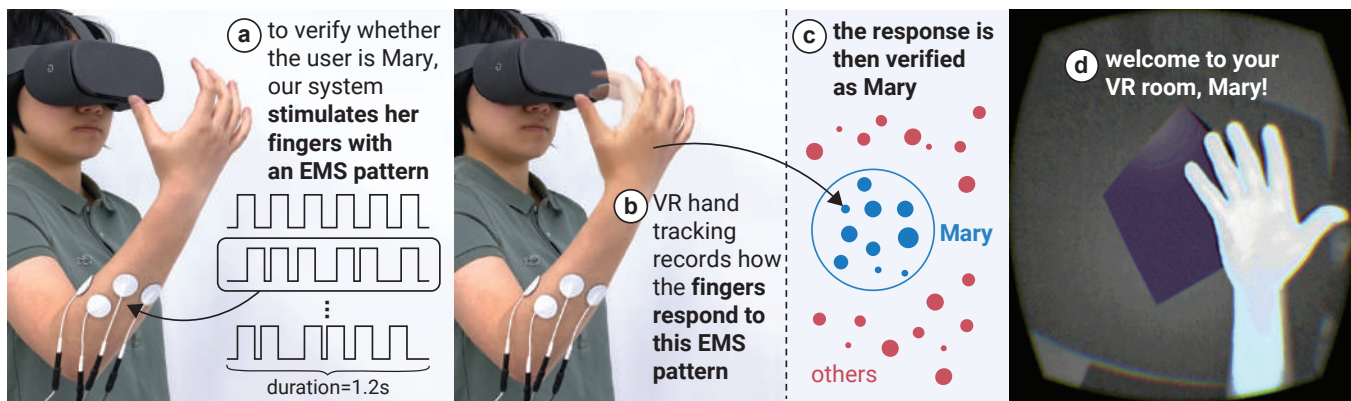


Figure 1: We propose a novel modality for authentication: electrical muscle stimulation (EMS). To explore it, we created an interactive system that (a) stimulates the user’s forearm muscles with electrical impulses (i.e., using one of 68M possible EMS challenges); (b) measures the user’s involuntary finger movements, which are unique because everybody’s physiology is different; (c) verifies this response using an authentication model, and immediately eliminates this challenge, making our system secure against data breaches and replay attacks as it never reuses the same challenge. We demonstrate it here using the example of (d) authenticating a VR user without passwords or PINs.

ABSTRACT

We propose a novel modality for active biometric authentication: electrical muscle stimulation (EMS). To explore this, we engineered an interactive system, which we call ElectricAuth, that stimulates the user’s forearm muscles with a sequence of electrical impulses (i.e., EMS challenge) and measures the user’s involuntary finger movements (i.e., response to the challenge). ElectricAuth leverages EMS’s *intersubject variability*, where the same electrical stimulation results in different movements in different users because everybody’s physiology is unique (e.g., differences in bone and muscular structure, skin resistance and composition, etc.). As such, ElectricAuth allows users to login without memorizing passwords or PINs. ElectricAuth’s challenge-response structure makes it secure

against data breaches and replay attacks, a major vulnerability facing today’s biometrics such as facial recognition and fingerprints. Furthermore, ElectricAuth never reuses the same challenge twice in authentications – in just one second of stimulation it encodes one of 68M possible challenges.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*; • **Security and privacy** → **Authentication**; *Biometrics*.

KEYWORDS

electrical muscle stimulation, biometric authentication, wearable

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1 INTRODUCTION

Biometric authentication is a technique that identifies an individual by their unique biological characteristics, such as their iris, fingerprints, or even one's voice. To identify their users, these interactive systems compare a previously stored biometric key to incoming, typically real-time, biometric data of the user wishing to authenticate. Compared to traditional password or PIN based systems, biometric authentication offers significantly better usability as it does not require users to memorize passwords or PINs. As such, biometric authentication is getting widely adopted, replacing passwords in many contexts.

However, the key feature of biometric authentication is typically also its key flaw: once the biometric data is compromised (e.g., stolen in database breaches or recorded by an external attacker), there is nothing the user can do to securely re-use their own data. For example, if someone steals a user's fingerprints, this user can never trust a fingerprint-based interactive system. Unfortunately, these threats are not theoretical and many biometric systems have been breached. For instance, the biggest known biometric data breach involved a database of 27.8M records, including fingerprints and faces [8].

To tackle this shortcoming, researchers turned to interactive systems that feature a *challenge-response* as a form of active biometric authentication. One example is Velody [12], which challenges a user by vibrating her palm and measuring the user's unique vibration-response. The advantage of these systems is that, if the stored challenge-response pairs are breached, the system can quickly recover by simply asking the user to submit responses to a new set of challenges. As such, researchers seek to find more modalities that afford challenge-response biometric authentication.

In this work, we explore a novel modality for active biometrics: electrical muscle stimulation (EMS). To understand and evaluate the potential of EMS as a biometrics system for interactive applications, we engineered a prototype that performs user authentication via EMS. Our system, which we call ElectricAuth, stimulates the wearer's forearm muscles with an EMS-based challenge, i.e., a 1.2s long sequence of electrical impulses on four of the user's muscles. Then, it measures the user's involuntary movements that result from this EMS challenge. In Figure 1, we illustrate our system with the example of authenticating a user in VR. Here, ElectricAuth uses the VR headset's hand tracking to observe the response of the user's muscles to the EMS-challenge as their individual finger muscles are actuated.

ElectricAuth makes two key contributions in the design of EMS-based active biometric authentication: (1) ElectricAuth authenticates users by leveraging what is typically seen as the biggest disadvantage of EMS: *intersubject variability*, i.e., the same electrical stimulation results in different movements in different users because everybody's physiology is different [3, 10]. (2) ElectricAuth generates a very large pool of challenges by exploring an underutilized property of EMS: *temporal dependency*, i.e., muscles respond differently depending on their current state of contraction, which can be altered by varying the timing between two impulses. As such, ElectricAuth is robust against data breaches and replay attacks because it never reuses the same challenge twice in authentications.

2 RELATED WORK

The work presented in this paper builds on the fields of wearables, electrical muscle stimulation, and biometrics.

2.1 Electrical muscle stimulation

Electrical muscle stimulation (EMS) is a technique from medical rehabilitation [23] that induces involuntary movements by delivering electrical impulses to the user's muscles. This is typically achieved by non-invasive methods such as attaching pairs of electrodes to the user's skin (e.g., on top of the muscles that control finger movement, located in the forearm). Electrode pairs are typically driven using safe and medical compliant muscle stimulators [11].

Recently, researchers started to engineer interactive devices based on EMS. These tend to fall into two broad categories: (1) haptic devices that increase immersion/realism of virtual environments, and (2) interactive devices that facilitate information access via proprioception. As far as interactive devices that increase immersion, EMS has been used to render forces in mobile devices [13], virtual reality [14, 17] or augmented reality [6, 18]. As a means of general information access, EMS has been especially used for haptic training (e.g., learning a musical instrument [25], operating a tool the user is not familiar with [16]) or eyes-free communication (e.g., communicating walking directions via leg stimulation [24], communicating a state of a variable via wrist movements [15]).

Unlike these interactive systems that use EMS as a form of force feedback or as an information channel, we explore EMS in a new direction: leveraging user's unique muscular responses to EMS as a form of active biometric authentication.

2.2 Biometric authentication

Biometric authentication is designed to authenticate or verify an individual by their unique biological characteristics. Existing biometric systems can be categorized into two types: passive and active. Passive biometrics rely on physiological characteristics that naturally occur in users (e.g., fingerprints). Conversely, active biometrics are inherently interactive as they inject stimuli and perform the authentication by measuring users' physiological responses to the stimuli.

Passive biometrics are particularly vulnerable to data thefts and replay attacks as reported by numerous incidents and studies [1, 5, 9, 20, 26–28]. This is because the identity (also known as "key") associated with each user is physically "hard-coded" and is used *repeatedly* for all authentications. Moreover, in most passive biometric systems, there is a small number of available biological traits per user that act as suitable keys, e.g., once all ten fingerprints are compromised, this user can never again rely on interactive systems based on fingerprint authentication.

Compared to passive biometrics, active systems are robust against data thefts and replay attacks. This is because each user can potentially generate many challenges, each triggering a different response. The system uses a new challenge in each authentication, preventing attackers from using previously observed responses to breach it.

Our novel exploration of EMS-based authentication provides the advantages of movement-based challenge-response while automatically delivering the challenge and eliciting the user's *involuntary* response.

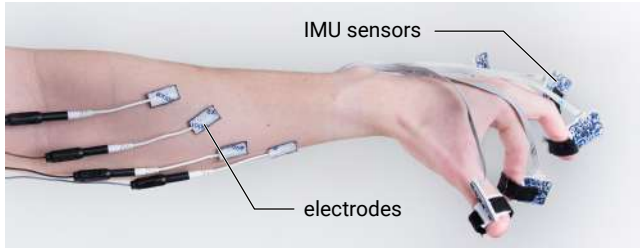


Figure 2: Our prototype consists of three components: (1) a EMS device; (2) a motion sensor and (3) a trained machine learning model (not shown in the figure).

3 IMPLEMENTATION

Our prototype¹, which is depicted in Figure 2, consists of three components: (1) a medically-compliant **EMS device** that delivers EMS challenges to the user, (2) a **motion sensor** that captures the actuated limb’s movements, and (3) a **trained machine learning model** that classifies the user’s movements and performs authentication.

1. EMS hardware. For delivering EMS impulses we use the Hasomed Rehaslim, a medical compliant device with eight individually controllable EMS channels. This device has often been used in interactive systems based on EMS [17–19]. To control the EMS stimulation, our software sends serial commands via USB using the Hasomed’s Science Protocol [7]. These impulses have a latency of <1ms.

As with any device based on EMS, we start by calibrating the electrode placement for each user at her registration session. Our calibration aims at targeting four muscles on the user’s forearm that actuate finger and wrist rotation. At the anterior forearm we stimulate two muscle groups: (1) primarily the *flexor carpi radialis* and partially the *flexor digitorum profundus*; and, (2) the *flexor pollicis longus*. At the posterior forearm we stimulate two muscle groups: (1) primarily the *extensor digitorum* and partially the *extensor digiti minimi*, *extensor pollicis brevis & longus*; and, (2) the *extensor indicis*. As is typical with EMS-based systems, these electrode positions are adjusted for each user during the registration session to ensure comfort. After calibration, the resulting electrode layout for a particular user is fixed by making an EMS-electrode sleeve (fabric with electrodes stitched to it) that this user wears any time they use ElectricAuth.

Our EMS stimuli on all electrode locations are the same: single-shot square-impulses with an intensity of 10mA and a pulse-width of 200μs. EMS challenges are constructed by sequencing these standardized pulses to one of the four channels the user’s forearm is connected to. For instance, one can construct a challenge with six impulses, each followed by a resting period.

To significantly increase our challenge pool, we explored a rather unused property of human muscles that causes them to respond differently to EMS depending on their current state of contraction. We call this *temporal dependency*. The use of temporal dependency affords a large EMS challenge pool by varying the time gaps between

consecutive stimuli. For our ElectricAuth prototype, we utilize a sequence of 6 pulses, 4 EMS channels and 7 different time gaps ($\tau = \frac{1}{30}s, \frac{2}{30}s, \dots, \frac{7}{30}s$), which in early pilots we found to lead to sufficiently different movement outcomes. The resulting maximum number of unique challenges is 68, 841, 472 (68M).

2. Motion sensing. We utilized a set of five 9-DOF inertial measurement units, attached to the fingers via a 3D printed ring (NXP Precision 9DoF, comprised of the FXOS8700 3-Axis accelerometer and magnetometer and the FXAS21002 3-axis gyroscope). These sample the fingers’ acceleration and rotation at 50Hz (post-sample interpolated to 100Hz) with a precision of ±4g at 14-bit for acceleration and ±250°/s at 16-bit for rotations; note that we do not use the magnetometer. These IMUs are sampled by a ATSAMD21G18 ARM Cortex M0 48 MHz processor, via a TCA9548A I2C Multiplexer. Finally, our sensing board relays the IMU data via serial over USB to our software.

3. Authentication software and pipeline. The software component of ElectricAuth, written in Python, handles all the interactions between EMS device, motion sensing, model training and real-time authentication. The pipeline of ElectricAuth, which is depicted in Figure 3, is comprised of two phases: (a) **registration** and (b) **authentication**. In the **registration** phase, registering a new user (after calibration) is as follows: (1) a set of n EMS challenges are sent one at a time; (2) the user’s movements in response to each challenge are recorded; (3) these responses are used to train a machine learning-based authentication model for this user. In the **authentication** phase, verifying a user’s identity in run-time is as follows: (i) one random EMS challenge belong to the claimed identity is chosen, deleted immediately from the database, and sent to the user via EMS; (ii) the user’s movements in response to the challenge are recorded; (iii) the motion responses are fed into the trained authentication model of the claimed identity; (iv) the system determines whether this user is legitimate or not.

We implemented a robust authentication model that, for each registered user, integrates two deep neural network (DNN) models. Specifically, authentication starts with **(1) an unsupervised anomaly detector**, which verifies whether a response was produced by the user the model belongs to (i.e., the legitimate user); this step prevents *impersonation attacks*, in which a different user attempts to gain identity of the legitimate user. If a response passes the anomaly detector, it then enters **(2) a challenge classifier**, which detects and rejects replay attacks by verifying whether the response is the reaction to the challenge used in the current authentication session.

For our prototype, we apply a reconstruction error based anomaly detection system [22]. Specifically, we use variational autoencoder (VAE) [4], a DNN architecture well-known for *automatically* capturing complex patterns in target data. In addition, we use a Convolutional Neural Network (CNN) for challenge classification task [21]. Both models are trained using only the challenge-response pairs of this legitimate user collected during registration. When the user (re)registers a new set of challenges, we retrain both models from scratch using the new data.

¹More details can be found in our paper [2]. We also provide the source code at <http://sandlab.cs.uchicago.edu/electricauth>.

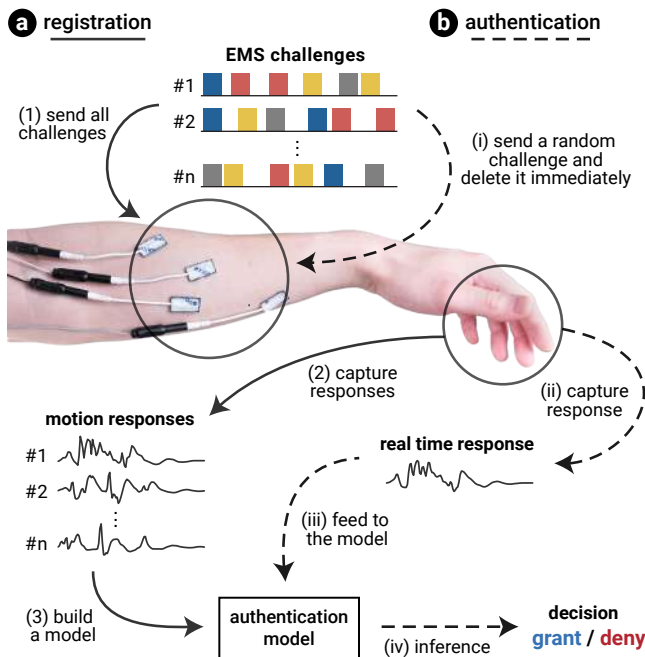


Figure 3: Interactive pipeline for the registration (registering a new user, marked by solid lines) and authentication phase (interactive use in runtime, marked by dashed lines).

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